Advancing Semiconductor Fabrication: A CNN-Based Wafer Defect Detection with XAI Insights

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***Abstract*—Semiconductor fabrication encounters substantial obstacles in defect identification, adversely affecting yield and quality of the product. Conventional manual classification techniques are ineffective, attaining accuracy rates below 45%. This study develops and evaluates deep learning classifiers, including ResNet50, a custom CNN, Inception v3, and a machine learning classifier SVM, utilizing the WM-811K dataset, which comprises nine distinct wafer defect classes, to improve classification accuracy and minimize class imbalance. ResNet50 had exceptional performance, attaining 99.93% accuracy for "Near-full" defects and 95.77% for "Local" defects. Furthermore, the incorporation of Gradient-weighted Class Activation Mapping (Grad-CAM), an Explainable Artificial Intelligence (XAI) technique, with ResNet50 enhanced model interpretability by emphasizing significant defect areas. These automated classification method provides a reliable and efficient approach for detecting defects, thereby optimizing the semiconductor manufacturing process and improving industry growth by enhancing yield and quality.**

***Index Terms*—CNN, Wafer Maps, ResNet50, Grad-CAM, XAI**

I. Introduction

The semiconductor industry is expanding rapidly as a result of the continuous development of technology, which enables the design and manufacturing devices that are both more precise and high-performing. Despite these advancements, semiconductor manufacturing remains to be a difficult, expensive, and time-intensive process, involving several chemical, mechanical, and electrical procedures including deposition, etching, photolithography, chemical planarization, ion implantation, and diffusion [1]. The fabrication process begins with the production of integrated circuits (ICs) by constructing circuit architectures spanning several wafer layers, which are then connected with wires. Maintaining the cleanliness of the wafer surface and the exact alignment of all circuit layers is essential for the production of high-quality integrated circuits. Regardless of the presence of highly skilled engineers and cutting-edge automated machinery functioning in clean surroundings, the fabrication of completely defect-free wafer dies remains unattainable [2]. Defects in semiconductor wafers may occur at any phase of the fabrication process, frequently caused by human errors, particulate contamination from machinery, chemical residues, and various other causes [3]. These defects have a negative impact on wafer quality and overall reliability of the product, therefore effective detection and classification of the defects are crucial for optimizing the fabrication process and fulfilling market needs. Wafer maps, a two-dimensional graphical representations of semiconductor wafers, are essential for visualizing and analyzing defects. They provide critical information about the location, dimensions, and characteristics of defects, facilitating the detection of unusual patterns that reveal fundamental challenges in the fabrication process [4]. Experienced process engineers generally classify wafer defects into distinct pattern categories, such as Center, Donut, Local, Edge-Local, Edge-Ring, Scratch, Random, Near-Full, and None [5]. Each defect pattern correlates with particular anomalies in the fabrication process. For instance, Scratch defects originate from mechanical handling issues, Center defects result from uniformity problems in chemical and mechanical planarization, and Donut defects occur owing to unequal doping concentrations [6]. Traditionally, defect detection and classification have relied on manual examinations conducted by engineers utilizing high-resolution microscopes to assess defect characteristics. This manual method is inefficient, expensive, and time-intensive, frequently yielding accuracy rates below 45% [7]. The underlying limitations of manual defect identification highlight the necessity for automated classification methods that ensure reliability and accuracy in identifying wafer defects. A range of machine learning (ML) methodologies has been developed to enhance the efficiency and accuracy of wafer map defect class detection. Convolutional neural networks (CNNs) have been utilized to improve the precision and accuracy of defect pattern recognition [3]. Furthermore, ensemble approaches and clustering algorithms, including DBSCAN, have been examined to enhance the detection and classification frameworks for wafer bin maps [6]. Kyeong et al. [8] presented a CNN-based framework featuring distinct models for each defect type, facilitating precise classification of mixed-type patterns in wafer bin maps, even under noisy settings. Cheon et al. [1] introduced a CNN-based approach for the automatic

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classification of wafer surface defects, incorporating a k-nearest neighbors (k-NN) technique to recognize previously unobserved defect categories, attaining a classification accuracy of 96.2%. Zheng et al. [9] conducted a study that developed a deep learning-based convolutional neural network (DCNN) model for the automated classification of wafer map failure patterns. Their methodology surpassed most established machine learning models and deep learning architectures for accuracy, while it encountered challenges in identifying the 'Local' failure pattern type. Another study by Saqlain et al. [10] demonstrated a deep layered CNN-based model (CNN-WDI) for the identification of wafer defects in semiconductor manufacturing, employing the WM-811K dataset comprising nine defect categories. As circuit integration density escalates and wafer designs become increasingly complex, the diversity of defects expands. This development highlights the need for accurate and effective defect classification techniques. This study develops and evaluates deep learning methodologies for the classification of semiconductor wafer defects, with the objective to develop high-performance models that effectively differentiate traditionally difficult defect categories while minimizing dataset class imbalance. This research evaluates and compares the performance of four classifiers: ResNet50, a traditional Convolutional Neural Network (CNN), Inception v3, and Support Vector Machine (SVM) in the context of semiconductor wafer defect detection. Utilizing the publicly available WM-811K dataset, we systematically preprocess and enhance the data to solve the issues of class imbalance and irregular image dimensions. The classifiers are subsequently trained and validated utilizing metrics including accuracy, precision, recall, and F1-score to determine their efficiency. Our methodology integrates Gradient-weighted Class Activation Mapping (Grad-CAM), an Explainable Artificial Intelligence (XAI) technique, with the ResNet50 model. This approach enhances model interpretability by emphasizing areas within wafer maps that affect classification, thus validating the accuracy and reliability of predictions.

II. Methodology

A. Dataset Description

The WM-811K dataset is a comprehensive, open-access collection of semiconductor wafer map images, consisting of a total of 811,457 wafer maps that were collected from 47,543 production lots in a real-world semiconductor fabrication environment. In optimal conditions, each lot would produce 25 wafer maps, culminating in a total of 1,157,325 wafers. Nonetheless, complications such as sensor malfunctions or other unidentified factors diminished the actual count of wafer mappings to 811,457. Each wafer map is accompanied by supplementary metadata, encompassing wafer die size, lot name, wafer index, and expert-defined failure category labels. Approximately 21.3% of the total dataset, or 172,950 wafer maps, include labels, while 78.7% do not. The "None" category predominates within this labeled group, comprising 147,431 wafer mappings and resulting in a major class imbalance. Only 25,519 wafer maps, including 3.1% of the total dataset and 14.8% of the labeled segment, illustrate actual failure patterns. The distribution of these defect classes is as follows: Center (4,294; 2.5%), Donut (555; 0.3%), Edge-Loc (5,189; 3.0%), Edge-Ring (9,680; 5.6%), Local (3,593; 2.1%), Random (866; 0.5%), Scratch (1,193; 0.7%), and Near-Full (149; 0.1%). Figure 1 illustrates examples of wafer map images, each corresponding to one of the eight specified failure types.

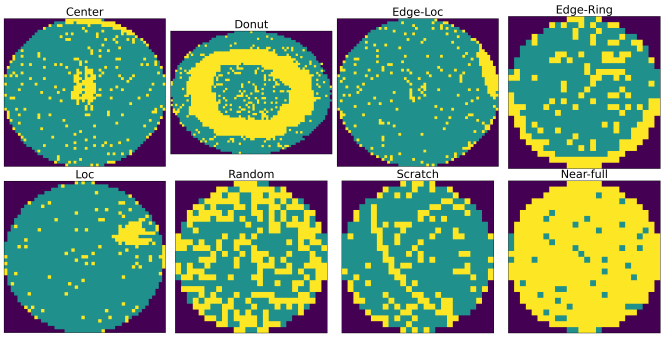


Fig. 1: Examples of eight distinct wafer defect classes.

B. Data preprocessing

The raw WM-811K dataset presents multiple issues that require a careful data preprocessing approach. The wafer map images were initially displayed in 346 different dimensions, which made direct neural network input impracticable. To standardize input dimensions, all images were resized to the dataset's median dimensions of 53×52 pixels, a roughly square configuration already found within the original image collection. Before executing this uniform restructuring, images exhibiting highly rectangular proportions, characterized by one dimension being at least 50% greater than the other, were removed in order to ensure consistency in wafer geometry. Alongside dimensional normalization, a quality check was conducted to detect and eliminate problematic images. For each image, the ratio of background pixels to non-background pixels which represent chip regions was calculated. Those having a non-background ratio less than 0.6 were removed because they were considered flawed. Once these abnormalities and the excessively shaped images were removed, 170,246 images were left. At this point, only the necessary pixel data and associated defect class labels remained after unnecessary metadata columns (dieSize, lotName, waferIndex, trainTestLabel, and shape) were removed. A three-level encoding approach was used to encode the pixel values in the retained images: 0 for the background, 1 for normal chips, and 2 for defective chips. Even after thorough cleaning and normalization, the dataset still showed a clear class imbalance, with most classified samples falling into the "None" category (no defects). In order to guarantee that minority classes are fairly represented throughout training, this imbalance calls for the application of data augmentation techniques in later modeling stages. The improved dataset was then divided into

training and test sets, with 20% put aside for evaluation and 80% of the images used for training.

C. Data Augmentation

To prepare the WM-811K dataset for model training, several data augmentation techniques were implemented to normalize input dimensions and resolve class imbalances. The wafer map images, displaying 346 unique dimensions, were resized to a standardized format of 224×224 pixels to ensure compatibility with our proposed ResNet50 CNN model. The resizing was accomplished by employing a padding function that incorporated background pixels, thus preserving the visual integrity of the wafer patterns without modifying their look. Both the training and test datasets underwent a padding procedure, employing nearest-neighbor interpolation to maintain the original pixel encoding: 0 for background, 1 for normal chips, and 2 for anomalous chips. Following resizing, the anomalous images underwent geometric changes to improve minority classes and lessen class imbalance. Each anomalous wafer map underwent horizontal and vertical flips, along with rotations of 90° and 270°, thereby augmenting the representation of minority defect classes. After undergoing these changes, the images were transformed from grayscale to RGB format by dividing the single channel into three channels, and they were normalized by scaling pixel intensities to the [0,255] range. This modification made sure that the CNN could use three-channel pictures with dimensions of 224, 224, and 3 as input. The augmented and normalized images were systematically stored in specified training and test directories, categorized into subfolders for each defect class. Even with these augmentation efforts, the dataset was still quite unbalanced, with 116,022 images in the "None" class and far less in the other defect types. To balance the dataset, a specific number of randomly selected images from highly imbalanced defect classes, including the "None" category, were removed from both the training and test sets. In order to guarantee fair representation across all nine defect categories and improve the model's capacity to generalize and correctly categorize minority classes, this reduction was put into place. Lastly, TensorFlow's image-dataset-from-directory function was used to load the balanced dataset. This function created batched datasets appropriate for model training and evaluation by inferring labels from the directory structure. The final dataset included 26,623 test images and 106,492 training images, evenly split among the nine defect classes.

D. ResNet50 CNN Architecture

We utilized a Residual Network (ResNet50) architecture, recognized for its effectiveness in image identification, to precisely categorize wafer map anomalies in the WM-811K dataset. The ResNet50 model was initialized using weights pre-trained on the ImageNet dataset, excluding the top classification layers to enable customization for our specific anomaly detection issue. This foundational model is consistent with the standardized dimensions that were established during data preprocessing, accepting input images of size $224 \times 224 \times 3$. The architecture was augmented with custom classification layers to adapt ResNet50 for the nine-class anomaly classification task. Following the base model, a Global Average Pooling (GAP) layer was introduced to reduce the spatial dimensions of the feature maps, effectively summarizing each feature map into a single value:

$$\text{GAP}(x) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_{i,j} \tag{1}$$

where H and W represent the height and width of the feature maps, respectively. This operation transforms the feature maps into a fixed-size vector, irrespective of the input dimensions, thereby facilitating subsequent processing.

Subsequently, a Batch Normalization (BN) layer was incorporated to stabilize and accelerate the training process by normalizing the output of the preceding layer:

$$\text{BN}(x) = \gamma \left(\frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \right) + \beta \tag{2}$$

where μ and σ^2 denote the mean and variance of the batch, and γ and β are learnable parameters. This normalization mitigates internal covariate shift, enhancing convergence rates and overall model performance.

To mitigate overfitting, Dropout layers with a dropout rate of 0.3 were employed, randomly deactivating a fraction of neurons during training:

$$\text{Dropout}(x) = \begin{cases} \frac{x}{1-p} & \text{with probability } 1-p \\ 0 & \text{with probability } p \end{cases} \tag{3}$$

where $p = 0.3$ is the dropout rate. This regularization technique prevents the model from becoming overly reliant on specific neurons, promoting the development of more robust feature representations.

A Dense layer with 1,024 units and ReLU activation was added, coupled with L2 regularization to further prevent overfitting:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{categorical_crossentropy}} + \lambda \sum_i W_i^2 \tag{4}$$

where $\lambda = 0.01$ is the regularization parameter and W_i represents the weights of the Dense layer. This penalization of large weights encourages the model to maintain smaller, more generalized weight values. An additional Dropout layer with a rate of 0.3 was incorporated to reinforce regularization, followed by an output Dense layer with nine units and softmax activation:

$$\hat{y}_c = \frac{e^{z_c}}{\sum_{k=1}^{C} e^{z_k}} \tag{5}$$

where z_c is the input to the c-th output neuron and $C = 9$ is the number of anomaly classes. This softmax activation facilitates probabilistic classification, enabling the model to assign a probability distribution over the nine classes for each input image. The complete model was instantiated using Keras' Model API, defining the inputs as those of the base model and

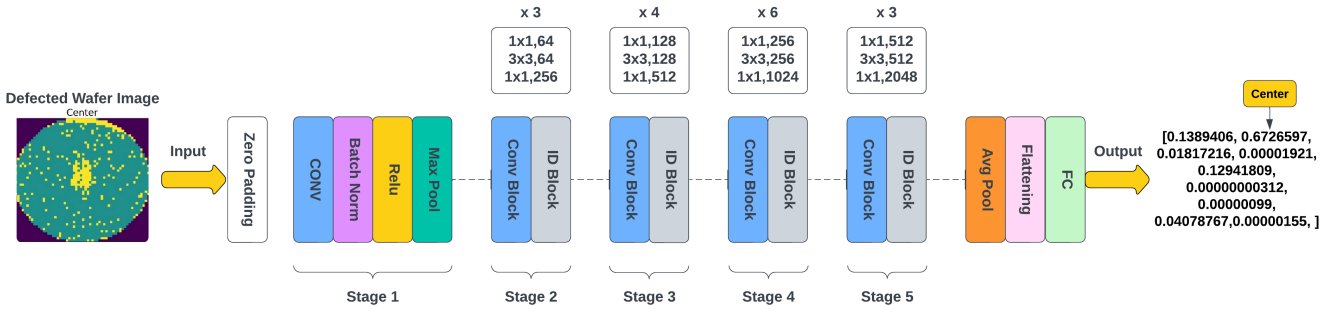


Fig. 2: Proposed ResNet-50 CNN model architecture.

the outputs as the predictions from the custom classification layers. The model was compiled with the Adam optimizer, characterized by its adaptive learning rate capabilities:

$$\theta_{t+1} = \theta_t - \frac{\eta \hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \tag{6}$$

where $\eta = 0.001$ is the learning rate, $\hat{m}_t$ and $\hat{v}_t$ are the bias-corrected first and second moment estimates of the gradients, and ϵ is a small constant to prevent division by zero. The loss function employed was categorical cross-entropy, defined as:

$$\mathcal{L} = -\sum_{c=1}^{C} y_c \log(\hat{y}_c) \tag{7}$$

where y_c is the true label and $\hat{y}_c$ is the predicted probability for class c. To address the significant class imbalance inherent in the dataset, class weights were computed to balance the contribution of each class during training. These weights w_c were inversely proportional to the class frequencies:

$$w_c = \frac{N}{C \times N_c} \tag{8}$$

where N is the total number of samples, C is the number of classes, and N_c is the number of samples in class c. In order to keep the model from being biased in favor of the majority "None" class, this weighting strategy makes sure that minority classes are given proportionately more attention during the optimization process. The training procedure was conducted in two stages to enhance model performance and prevent overfitting. Initially, the custom classification layers were trained for 10 epochs while the layers of the ResNet50 base model remained frozen, enabling the new layers to interpret the pre-extracted high-level features without altering the base weights. The training was monitored using several callbacks: Early Stopping, Model Checkpoint and ReduceLROnPlateau. In the subsequent fine-tuning phase, the last fifteen layers of ResNet50 were unfrozen to adapt to the specific wafer map data, and the model was recompiled with a reduced learning rate of 0.0001 to ensure that the pre-trained weights were delicately updated without disrupting the established feature hierarchies. Fine-tuning was conducted for an additional 15 epochs under the same set of callbacks and class weighting scheme. The final architecture comprised approximately 25.7 million parameters, of which 7.6 million were trainable and 18.1 million remained non-trainable due to the frozen base layers.

E. Experimental Setup

The experiments were conducted on a personal computer equipped with an AMD Ryzen 7 5800X3D 3.40 GHz processor, 32 GB of RAM, and an NVIDIA GeForce RTX 3080 GPU with 10 GB of memory. The ResNet50 model was developed using TensorFlow (2.10.0), Keras, and NumPy (1.23.5) libraries within a Jupyter Notebook environment, utilizing Python (3.10.11).

III. Result Analysis

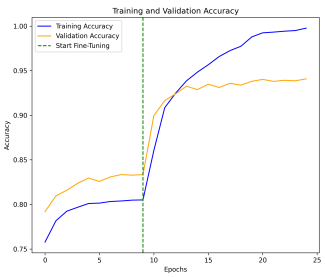
A. ResNet50

The ResNet50 CNN model showed strong generalization and discriminative abilities on the WM-811K wafer map dataset, as indicated by the performance curves, metrics, and visual assessments. After the initial phase of training, the initiation of fine-tuning at approximately the tenth epoch significantly enhanced the accuracy of both training and validation, as illustrated in Figure 3a. Additionally, the loss values in both sets were reduced, as illustrated in Figure 3b. The refined model achieved near-perfect classification performance for several categories, including "Near-full" anomalies, which achieved an accuracy of 99.93% and an F1-score of 0.9970.

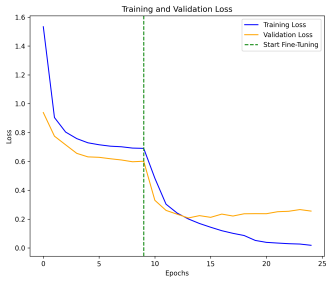
TABLE I: Per-class Performance Metrics of the ResNet-50 CNN model

Class	Accuracy	Precision	Recall	F1-score
Center	0.9859	0.9346	0.9414	0.9380
Donut	0.9953	0.9758	0.9830	0.9794
Edge-Loc	0.9677	0.8562	0.8577	0.8570
Edge-Ring	0.9938	0.9671	0.9787	0.9728
Loc	0.9577	0.8472	0.7617	0.8022
Near-full	0.9993	0.9963	0.9977	0.9970
Random	0.9964	0.9837	0.9840	0.9838
Scratch	0.9846	0.9268	0.9373	0.9321
none	0.9843	0.8928	0.9542	0.9225

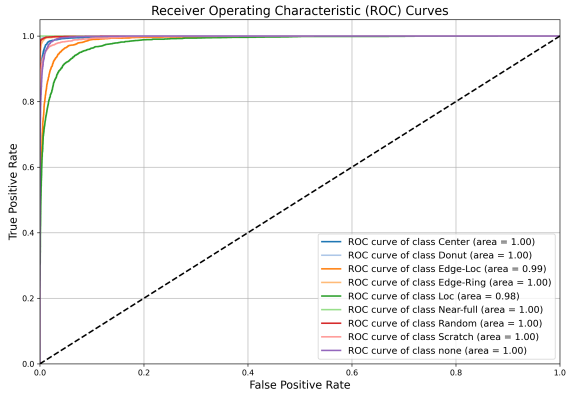
Even the more challenging classes, such as ”Loc,” maintained acceptable performance levels (Accuracy = 95.77%, F1-score = 0.8022), as summarized in Table I. Figure 3c illustrates the



(a) Training and Validation Accuracy of ResNet-50



(b) Training and Validation Loss of ResNet-50



(c) ROC Curve for ResNet-50

Fig. 3: Overview of ResNet-50 performance evaluation.

Receiver Operating Characteristic (ROC) curves, highlighting the model’s remarkable discriminative capacity, as several classes attain an area under the curve (AUC) approaching or equal to 1.00. The confusion matrix portrayed in Figure 4 illustrates that misclassifications are relatively uncommon, and when they do occur, they frequently occur among classes with similar visual defect patterns, such as Edge-Loc and Edge-Ring. These findings suggest that, despite the model’s overall performance, specific classes that exhibit similar characteristics may require more advanced feature extraction techniques or supplementary training strategies to improve their discriminative capabilities.

B. XAI-Grad-CAM Visualization

To improve the interpretability of the ResNet50 CNN model applied on the semiconductor wafer dataset, we implemented Gradient-weighted Class Activation Mapping (Grad-CAM) as an Explainable Artificial Intelligence (XAI) method. Grad-CAM was employed to visualize the areas inside wafer map images that substantially impacted the model’s classification of nine distinct defect categories. Figure 5 displays an extensive Grad-CAM visualization for a representative wafer map featuring a central flaw, demonstrating the transition from

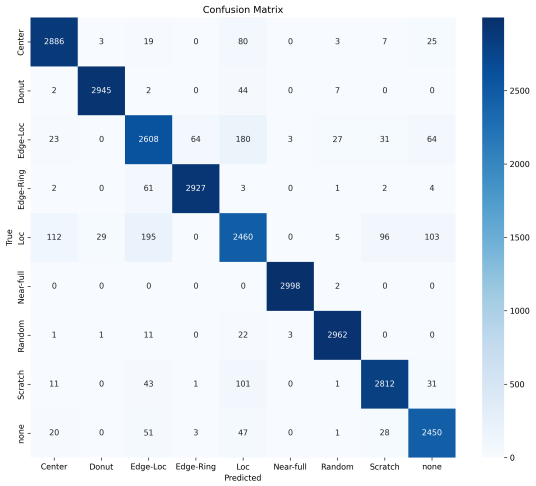


Fig. 4: Confusion Matrix for ResNet-50 CNN model.

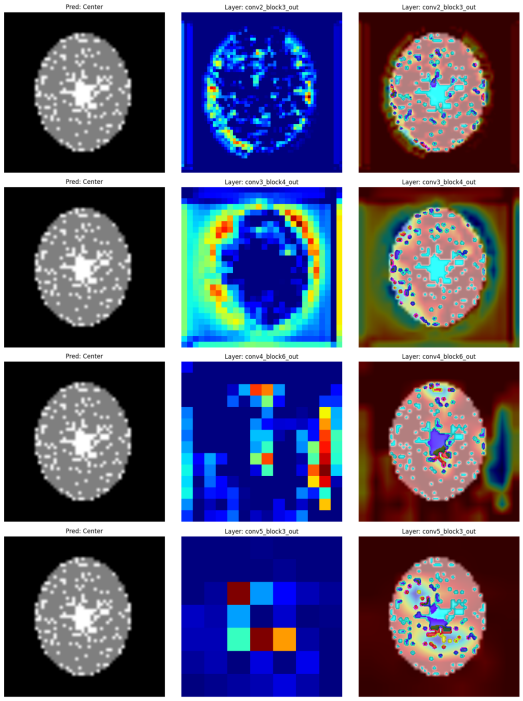


Fig. 5: Grad-CAM visualization of the ResNet50 CNN model.

the original wafer map to the Grad-CAM heatmap and the final overlay image. The original wafer map with a noticeable center defect is seen in the left panel. The central panel displays the Grad-CAM heatmap produced by the Final Convolutional Layer, illustrating the regions of the image that the model mostly concentrated on while detecting the central flaw. Initially, the Grad-CAM heatmap displayed broad activation regions in the Stage 2, Block 3 convolutional layer, capturing fundamental features such as edges and basic textures around the center defect. This laid the foundation for more complex feature extraction. Advancing to Stage 3, Block 4, the heatmap revealed a stronger concentration in the core area, signifying

the model's growing focus on relevant defect attributes while simultaneously incorporating contextual data from adjacent locations. In Stage 4, Block 6, the Grad-CAM visualization significantly improved, illustrating the precise boundaries and complex patterns of the central defect with greater clarity, indicating the model's advanced feature abstraction abilities. Lastly, the heatmap in the Stage 5, Block 3 layer demonstrated sharp and highly focused activations that were directly correlated with the center defect. This emphasized the model's capacity to identify and prioritize crucial regions that are essential for precise classification. The transition from Stage 2, Block 3 to Stage 5, Block 3 in Figure 5 highlights the hierarchical feature extraction characteristic of ResNet50, where first layers detect fundamental visual elements and subsequent layers integrate this data to identify intricate defect patterns. The right panel integrates the original wafer map with the Grad-CAM heatmap, providing an overlay visualization that clearly illustrates the model's attention mechanism. This overlay verifies that the ResNet50 model accurately focuses on the defective region, enhancing the credibility of its predicting skills. The concentrated activation at the center corresponds with the expert-defined defect region, demonstrating that the model not only identifies the defect but also correctly relates it to its geographical setting within the wafer map.

C. Comparative Analysis

In a comparative evaluation of four classification models applied to the WM-811K wafer defect dataset, as described in Table II, ResNet50 exhibited the greatest overall efficacy. It attained an accuracy of 94.08%, with precision, recall, and F1-score metrics all exceeding 0.94, thus outperforming Inception v3, custom CNN, and SVM. Although Inception v3 (accuracy: 88.16%) and the custom CNN (accuracy: 86.21%) produced rather strong outcomes, they continuously fell short of ResNet50 across all parameters. The SVM exhibited inferior performance compared to the deep learning methodologies, achieving an accuracy of 82.30% and an F1-score of 0.8118, possibly attributable to its dependence on manually crafted features instead of hierarchical representations derived directly from the data. These results emphasize the superior classification capabilities of advanced deep convolutional architectures, notably ResNet50, in the detection of semiconductor wafer defects.

TABLE II: Comparative analysis of the proposed models

Classifier	Accuracy	Precision	Recall	F1 Score
ResNet50	0.9408	0.9403	0.9409	0.9405
Custom-CNN	0.8621	0.8758	0.8641	0.8624
Inception v3	0.8816	0.8842	0.8822	0.8783
SVM	0.8230	0.8182	0.8076	0.8118

IV. Conclusion

This research investigated the utilization of deep learning models for the categorization of semiconductor wafer defects using the WM-811K dataset. With an accuracy of 94.08% and constant superiority over the other techniques in precision, recall, and F1-score, ResNet50 was the most effective of the four classifiers studied. The interpretability of the model's decision-making process was improved by the integration of explainable AI (XAI) with Grad-CAM, which provided valuable insights. A significant improvement in the training and evaluation outcomes of the model was achieved by addressing dataset challenges such as class imbalance and diverse image dimensions through effective preprocessing and augmentation techniques. In order to improve the performance of anomaly detection, future research could explore alternative model architectures, implement targeted training strategies, or employ advanced data augmentation techniques to better the detection of the "Loc" defect class on wafers. The findings underscore the potential of ResNet50, in conjunction with AI tools that are comprehensible, to serve as a reliable solution for the identification and classification of wafer defects. Consequently, they contribute to the development of automated and dependable quality control systems in semiconductor manufacturing.

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